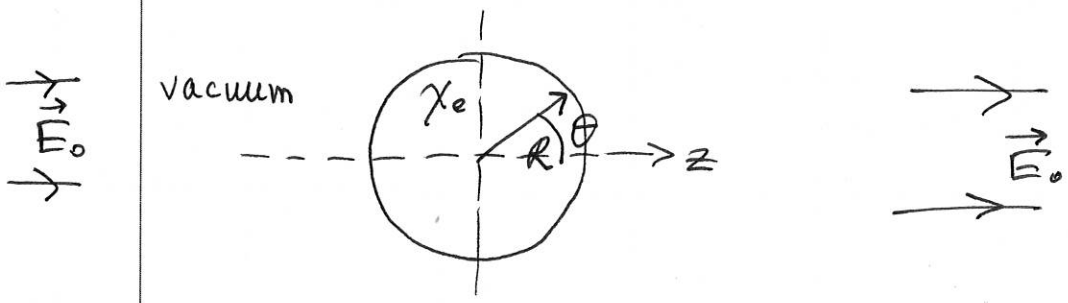
P1 Static situation, no external charges.

$$\vec{\nabla}\times\vec{E}=0$$

$$\vec{\nabla}\cdot\vec{D}=0 \qquad \vec{P}=\varepsilon_0\chi_e\vec{E} \Rightarrow \vec{D}=\varepsilon_0\vec{E}+\vec{P}=\varepsilon_0(1+\chi_e)\vec{E}$$

$$=\varepsilon_0\varepsilon_r\vec{E}=$$

$$=\varepsilon\vec{E}.$$



Hence $\vec{E}=-\vec{\nabla}\phi$ (so that $\vec{\nabla}\times\vec{E}=0$ is satisfied)

and $\vec{\nabla}\cdot\vec{D}=-\vec{\nabla}\cdot(\varepsilon_0\varepsilon_r\vec{E})=-\varepsilon_0\varepsilon_r\nabla^2\phi=0$

in regions with <u>constant $\varepsilon_r=1+\chi_e$</u>

Discussion on pages 295-6 of the textbook shows that in the case of uniformly polarized sphere electric field <u>outside</u> is that of a dipole $\vec{p}$

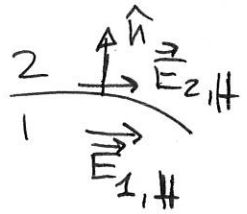
$$\vec{E}_{out}\sim\frac{3\hat{r}(\vec{p}\cdot\hat{r})-\vec{p}}{r^3}$$

while it is uniform inside $\vec{E}_{in}\sim\vec{p}$.

For linear dielectric $\vec{p}\sim\vec{E}_0$ so that we can try the following solution

$$\begin{cases}\vec{E}(\vec{r})=A\vec{E}_0+B\dfrac{3\hat{r}(\vec{E}_0\cdot\hat{r})-\vec{E}_0}{r^3}, & r>R\\ \vec{E}(\vec{r})=C\vec{E}_0, & r<R\end{cases}$$

Boundary conditions:



$\hat{n} = \hat{r}$

a) $\vec{E}_{1\parallel} = \vec{E}_{2\parallel}$

or, $(\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0.$

b) $E_{2,\perp} - E_{1,\perp} =$

$= (\vec{E}_2 - \vec{E}_1) \cdot \hat{n} = \dfrac{\sigma_f + \sigma_b}{\varepsilon_0}$

$\sigma_f = 0$ but $\sigma_b = \hat{P} \cdot \hat{n}$

But for $\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$ boundary condition is just

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = \sigma_f = 0$$

$$\vec{D} = \begin{cases} \varepsilon_0 \varepsilon_r \vec{E} & , r < R \\ \varepsilon_0 \vec{E} & , r > R \end{cases}$$

$\Rightarrow$ normal components: $\varepsilon_0 \varepsilon_r \vec{E} \cdot \hat{n} \Big|_{r \leq R} = \varepsilon_0 \vec{E} \cdot \hat{n} \Big|_{r \geq R}$

$$\Rightarrow C \varepsilon_r \vec{E}_0 \cdot \hat{r} = A \vec{E}_0 \cdot \hat{r} + B \frac{3 \hat{r} \cdot \hat{r} (\vec{E}_0 \cdot \hat{r}) - \vec{E}_0 \cdot \hat{r}}{R^3}$$

$$\Rightarrow C \varepsilon_r = A + \frac{2B}{R^3} \quad (1)$$

$\Rightarrow$ tangential components: $\vec{E} \times \hat{n} \Big|_{r \leq R} = \vec{E} \times \hat{n} \Big|_{r \geq R}$

$$C \vec{E}_0 \times \hat{r} = A \vec{E}_0 \times \hat{r} - \frac{B}{R^3} \vec{E}_0 \times \hat{r}$$

$$\Rightarrow C = A - \frac{B}{R^3} \quad (2)$$

Far from the sphere $\vec{E}(\vec{r}) \to \vec{E}_0 \Rightarrow A = 1$.

$\Rightarrow$ (1) + (2) gives: $(\varepsilon_r + 1)\, C = \frac{B}{R^3} + 2$

(1) − (2) gives: $(\varepsilon_r - 1)\, C = \frac{3B}{R^3}$

$$\Rightarrow (\varepsilon_r + 1)\, C = 2 + \frac{1}{3}(\varepsilon_r - 1)\, C$$

$$\left(\frac{2}{3}\varepsilon_r + \frac{4}{3}\right) C = 2 \quad , \quad C = \frac{6}{2\varepsilon_r + 4} = \frac{3}{\varepsilon_r + 2}$$

$$B = \frac{R^3}{3}(\varepsilon_r - 1)\frac{3}{\varepsilon_r + 2} = R^3 \frac{\varepsilon_r - 1}{\varepsilon_r + 2}$$

Or, in terms of $\chi_e = \varepsilon_r - 1$:

$$C = \frac{3}{3 + \chi_e} \quad , \quad \frac{B}{R^3} = \frac{\chi_e}{3 + \chi_e}$$

<u>Solution</u>

$r > R$: $\vec{E}(\vec{r}) = \vec{E}_0 + \frac{\chi_e}{3 + \chi_e} \frac{R^3}{r^3}\left(3\hat{r}\,(\vec{E}_0 \cdot \hat{r}) - \vec{E}_0\right)$

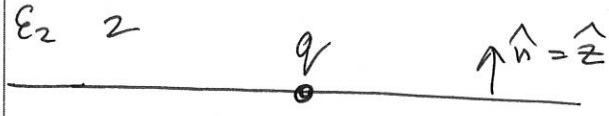
$r < R$: $\vec{E}(\vec{r}) = \frac{3}{3 + \chi_e}\vec{E}_0$

Now can find polarization inside the sphere:

$$\vec{P} = \varepsilon_0 \chi_e \vec{E} = \frac{3\chi_e \varepsilon_0}{3 + \chi_e}\vec{E}_0$$

Total dipole moment: $\vec{p} = \int \vec{P}\, d^3\vec{r} = \frac{4\pi}{3} R^3 \vec{P} = \frac{4\pi\varepsilon_0 \chi_e}{3 + \chi_e}\vec{E}_0 R^3$

P2 let charge q be at the origin.



let $\vec{E} \sim q\frac{\vec{r}}{r^3} = q\frac{\hat{r}}{r^2}$ $\qquad \vec{E} = \begin{cases} A_1 \frac{q\vec{r}}{r^3}, & \text{for region 1 } (z<0) \\ A_2 \frac{q\vec{r}}{r^3}, & \text{for region 2 } (z>0) \end{cases}$

(1) This choice ensures boundary condition for the normal components $(\vec{D}_2 - \vec{D}_1)\cdot\hat{n} = \sigma_f = 0$

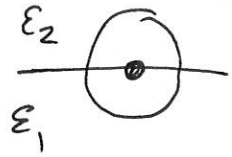
everywhere except for $\vec{r} = 0$ (where the charge is)

(2) Transverse components

$$(\vec{E}_2 - \vec{E}_1)\times\hat{n} = 0 \Rightarrow \frac{q}{r^3}(A_2 - A_1)\,\vec{r}\times\hat{z} = 0$$

$$\Rightarrow A_2 = A_1 = A.$$

Note that consideration in (1) avoids $\vec{r} = 0$ point where the charge is located. In terms of charge density $\rho_f = q\,\delta(\vec{r}) = q\,\delta(x)\,\delta(y)\,\delta(z)$ this is a very singular point.



However $\vec{\nabla}\cdot\vec{D} = \rho_f \Rightarrow \int \vec{\nabla}\cdot\vec{D}\,dV = \int \vec{D}\cdot d\vec{S} = \int \rho_f\,dV =$
$= q$

Because of the assumed spherical symmetry

$$\vec{D} = \vec{D}_1 = A\varepsilon_1 \frac{q\hat{r}}{r^2} \quad \text{for } z<0$$

$$\vec{D} = \vec{D}_2 = \varepsilon_2 A \frac{q\hat{r}}{r^2} \quad \text{for } z>0$$

$$\Rightarrow \int \vec{D}\cdot d\vec{S} = \varepsilon_1 A \frac{q\hat{r}}{r_0^2}\cdot 2\pi r_0^2 \hat{r} + \varepsilon_2 A \frac{q\hat{r}}{r_0^2}\cdot 2\pi r_0^2 \hat{r} = q$$

choose gaussian surface to be a sphere of radius r_0 $\Rightarrow$ area of a <u>hemisphere $= 2\pi r_0^2$</u>

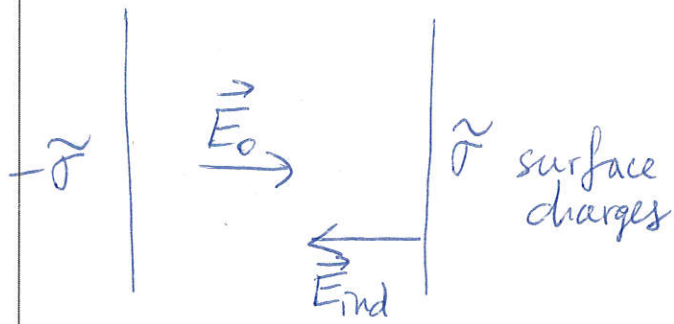
$$\Rightarrow 2\pi A q(\varepsilon_1+\varepsilon_2) = q \Rightarrow A = \frac{1}{2\pi(\varepsilon_1+\varepsilon_2)}$$

$$\Rightarrow \vec{D} = \begin{cases} \dfrac{\varepsilon_1 q}{2\pi(\varepsilon_1+\varepsilon_2)} \dfrac{\hat{r}}{r^2} & z<0 \\[2ex] \dfrac{\varepsilon_2 q}{2\pi(\varepsilon_1+\varepsilon_2)} \dfrac{\hat{r}}{r^2} & z>0 \end{cases}$$

$$\Rightarrow \vec{E} = \frac{q}{2\pi(\varepsilon_1+\varepsilon_2)} \frac{\hat{r}}{r^2} \quad \text{for all } \vec{r}.$$

$$\vec{E} = -\vec{\nabla}\varphi \Rightarrow \varphi(\vec{r}) = \frac{q}{2\pi(\varepsilon_1+\varepsilon_2) r}$$

P3



Surface charge $\tilde{\sigma}(t)$ produces induced electric field $\vec{E}_{ind}(t) = -\frac{\tilde{\sigma}(t)}{\varepsilon_0}$ acting against applied $\vec{E}_0(t)$.

Total electric field inside the metal

$$\vec{E}_{TOT}(t) = \vec{E}_0(t) - \frac{\tilde{\sigma}(t)}{\varepsilon_0} \qquad (1)$$

But charge build-up on the surface is due to the current $\vec{j}(t)$ in the metal:

$$j(t) \cdot \text{Area} = \frac{d(\tilde{\sigma} \cdot \text{Area})}{dt} = -i\omega\, \tilde{\sigma}(t)$$

let $\tilde{\sigma}(t) = \tilde{\sigma}_0 e^{-i\omega t} \Rightarrow \frac{d\tilde{\sigma}}{dt} = -i\omega\, \tilde{\sigma}_0 e^{-i\omega t} = -i\omega\, \tilde{\sigma}(t)$

Thus $\tilde{\sigma}(t) = \frac{j(t)}{-i\omega} = \frac{i\, j(t)}{\omega}$

Now we need to find $j(t) = n q\, v(t)$

Equation of motion $m\dot{\vec{v}} = q\vec{E}_{TOT} - \frac{m}{\tau}\vec{v}$, where $\vec{E}_{TOT}(t)$ is the local electric field = electric field experienced by the charge q.

let $\vec{E}_{TOT}(t) = \vec{E}\, e^{-i\omega t}$, $\vec{v}(t) = \vec{v}_0\, e^{-i\omega t}$

$$\Rightarrow \left(\frac{m}{\tau} - i\omega m\right) \vec{v}_0\, e^{-i\omega t} = q\, \vec{E}\, e^{-i\omega t}$$

$$\vec{v}(t) = \frac{q}{m\left(\frac{1}{\tau} - i\omega\right)} \vec{E}_{TOT}(t)$$

$$\vec{j}(t) = nq\, \vec{v}(t) = \frac{nq^2}{m\left(\frac{1}{\tau} - i\omega\right)} \vec{E}_{TOT}(t) = \frac{\varepsilon_0 \omega_0^2}{\frac{1}{\tau} - i\omega} \vec{E}_{TOT}(t)$$

$$= \sigma(\omega) \vec{E}_{TOT}\, e^{-i\omega t}$$

where $\omega_0^2 = \dfrac{nq^2}{m\varepsilon_0}$ = plasma frequency

Thus $\vec{E}_{TOT}(t) = \vec{E}_0(t) - \dfrac{i}{\varepsilon_0 \omega} \underbrace{\sigma(\omega) \vec{E}_{TOT}\, e^{-i\omega t}}_{j(t)}$

$$\left(1 + \frac{i\sigma}{\varepsilon_0 \omega}\right) \vec{E}_{TOT}\, e^{-i\omega t} = \vec{E}_0\, e^{-i\omega t}$$

$$\vec{E}_{TOT} = \frac{1}{1 + \frac{i\sigma}{\varepsilon_0 \omega}} \vec{E}_0 = \frac{1}{1 + \frac{i\omega_0^2}{\omega\left(\frac{1}{\tau} - i\omega\right)}} \vec{E}_0 =$$

$$= \frac{\omega\left(\frac{1}{\tau} - i\omega\right)}{\frac{\omega}{\tau} - i\omega^2 + i\omega_0^2} \vec{E}_0 = \frac{-i\,\omega\left(\omega + \frac{i}{\tau}\right)}{-i\left(\omega^2 + \frac{i\omega}{\tau} - \omega_0^2\right)} \vec{E}_0 =$$

$$= \frac{\omega\left(\omega + i/\tau\right)}{\omega\left(\omega + i/\tau\right) - \omega_0^2} \vec{E}_0 .$$

$$\vec{j} = qn\vec{v} = \frac{nq^2 \vec{E} e^{-i\omega t}}{m\left(\frac{1}{\tau} - i\omega\right)} = \frac{\varepsilon_0 \omega_0^2 \tau}{1 - i\omega\tau} \vec{E} e^{-i\omega t} =$$

$$= \sigma(\omega) \vec{E} e^{-i\omega t}$$

$$\sigma = \frac{\varepsilon_0 \omega_0^2 \tau}{1 - i\omega\tau}$$ Drude conductivity

Surface charge $\tilde{\sigma}$: $\frac{d\tilde{\sigma}}{dt} = \vec{j} \sim e^{-i\omega t}$

$$\tilde{\sigma} = \tilde{\sigma}_0 e^{-i\omega t}$$

$$\vec{j}(t) = \sigma(\omega) \vec{E} e^{-i\omega t} + \sigma(-\omega) \vec{E} e^{i\omega t}$$

$$j = \dot{\tilde{\sigma}} = -i\omega \tilde{\sigma}_0 e^{-i\omega t}$$

$$\tilde{\sigma}(t) = \tilde{\sigma}_0 e^{-i\omega t} = \frac{j(t)}{-i\omega} = \frac{i\,\sigma(\omega)}{\omega} e^{-i\omega t} \vec{E}$$

$$\vec{E} e^{-i\omega t} = \vec{E}_0 e^{-i\omega t} - \frac{\tilde{\sigma}}{\varepsilon_0} = \vec{E}_0 e^{-i\omega t} - \frac{i\,\sigma(\omega)}{\varepsilon_0 \omega} e^{-i\omega t} \vec{E}$$

$$\vec{E} e^{-i\omega t}\left(1 + \frac{i\,\sigma(\omega)}{\varepsilon_0 \omega}\right) = \vec{E}_0 e^{-i\omega t}$$

$$\vec{E} = \frac{1}{1 + \frac{i\,\sigma(\omega)}{\varepsilon_0 \omega}} \vec{E}_0$$

$$\vec{E} = \frac{1}{1 + \frac{i\omega_0^2\tau}{\omega(1-i\omega\tau)}} \vec{E}_0 = \frac{\omega(1-i\omega\tau)}{i\omega_0^2\tau - i\omega^2\tau + \omega} \vec{E}_0$$

$$= \frac{i\,\omega(1-i\omega\tau)}{\omega^2\tau + i\omega - \omega_0^2\tau} \vec{E}_0 =$$

$$= \frac{(\omega\tau + i)\omega}{\tau\left(\omega^2 + \frac{i\omega}{\tau} - \omega_0^2\right)} \vec{E}_0 = \frac{\omega\left(\omega + \frac{i}{\tau}\right)}{\omega\left(\omega + \frac{i}{\tau}\right) - \omega_0^2} \vec{E}_0$$

$$\vec{E}_{TOT}\, e^{-i\omega t} = \frac{\omega\left(\omega + \frac{i}{\tau}\right)}{\omega\left(\omega + \frac{i}{\tau}\right) - \omega_0^2} \vec{E}_0\, e^{-i\omega t}$$

$$\Rightarrow \vec{E}_{TOT}\, e^{i\omega t} = \frac{\omega\left(\omega - \frac{i}{\tau}\right)}{\omega\left(\omega - \frac{i}{\tau}\right) - \omega_0^2} \vec{E}_0\, e^{i\omega t}$$

in response to $\vec{E}_0 \cos(\omega t)$:

$$\vec{E}_{TOT}(t) = \frac{1}{2}\vec{E}_0 \left\{ \frac{\omega\left(\omega + \frac{i}{\tau}\right)}{\omega\left(\omega + \frac{i}{\tau}\right) - \omega_0^2} e^{-i\omega t} + \frac{\omega\left(\omega - \frac{i}{\tau}\right)}{\omega\left(\omega - \frac{i}{\tau}\right) - \omega_0^2} e^{i\omega t} \right\}$$

Apply $\vec{E}_0 \cos(\omega t) = \frac{1}{2} \vec{E}_0 \left(e^{i\omega t} + e^{-i\omega t}\right)$

$$\Rightarrow \vec{E}_{TOT}(t) = \frac{1}{2} \vec{E}_0 e^{-i\omega t} \frac{\omega(\omega + i/\tau)}{\omega(\omega + i/\tau) - \omega_0^2}$$

$$+ \frac{1}{2} \vec{E}_0 e^{i\omega t} \frac{-\omega(-\omega + i/\tau)}{-\omega(-\omega + i/\tau) - \omega_0^2} =$$

$$= \frac{1}{2} \vec{E}_0 \left(\frac{\omega(\omega + i/\tau)}{\omega(\omega + \frac{i}{\tau}) - \omega_0^2} e^{-i\omega t} + \frac{\omega(\omega - i/\tau)}{\omega(\omega - \frac{i}{\tau}) - \omega_0^2} e^{i\omega t} \right)$$

let's write $\vec{E}_{TOT}(t) = \vec{E}_1(\omega) e^{-i\omega t} + \vec{E}_2(\omega) e^{i\omega t}$

where $\vec{E}_{1,2}(\omega) = \frac{1}{2} \vec{E}_0 \frac{\omega(\omega \pm \frac{i}{\tau})}{\omega(\omega \pm \frac{i}{\tau}) - \omega_0^2}$

Then $\vec{j}(t) = \sigma(\omega) \vec{E}_1(\omega) e^{-i\omega t} + \sigma(-\omega) \vec{E}_2(\omega) e^{i\omega t}$

Drude Conductivity $\sigma(\omega) = \dfrac{\varepsilon_0 \omega_0^2}{\frac{1}{\tau} - i\omega}$

Dissipated power $P(\omega) = \frac{1}{T} \int_0^T dt \left(\vec{j}(t) \cdot \vec{E}_{TOT}(t) \right)$

$T = \frac{2\pi}{\omega}$ = period

$$P(\omega) = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} dt \left(\sigma(\omega)\,\vec{E}_1(\omega)\,e^{-i\omega t} + \sigma(-\omega)\,\vec{E}_2\,e^{i\omega t}\right)\cdot$$

$$\cdot\left(\vec{E}_1(\omega)\,e^{-i\omega t} + \vec{E}_2(\omega)\,e^{i\omega t}\right) =$$

$(x=\omega t)$

$$= \frac{1}{2\pi}\int_0^{2\pi} dx \Big(\sigma(\omega)\,\vec{E}_1(\omega)\cdot\vec{E}_2(\omega) + \sigma(-\omega)\,\vec{E}_2(\omega)\cdot\vec{E}_1(\omega)$$

$$+ \sigma(\omega)\,\vec{E}_1\cdot\vec{E}_1\,e^{-i2x} + \sigma(-\omega)\,\vec{E}_2\cdot\vec{E}_2\,e^{i2x}\Big)$$

But $\int_0^{2\pi} dx\, e^{\pm i2x} = 0$

$$\Rightarrow P(\omega) = \left(\sigma(\omega) + \sigma(-\omega)\right)\vec{E}_1(\omega)\cdot\vec{E}_2(\omega) =$$

$$= \varepsilon_0\omega_0^2\left(\frac{1}{\frac{1}{\tau}-i\omega} + \frac{1}{\frac{1}{\tau}+i\omega}\right)\left(\frac{1}{2}\vec{E}_0\cdot\frac{1}{2}\vec{E}_0\right)\frac{\omega(\omega+\frac{i}{\tau})}{\omega(\omega+\frac{i}{\tau})-\omega_0^2}\cdot$$

$$\cdot\frac{\omega(\omega-\frac{i}{\tau})}{\omega(\omega-\frac{i}{\tau})-\omega_0^2} = \frac{1}{4}E_0^2\,\frac{2\varepsilon_0\omega_0^2\frac{1}{\tau}}{\frac{1}{\tau^2}+\omega^2}\cdot$$

$$\cdot\frac{\omega^2\left(\omega^2+\frac{1}{\tau^2}\right)}{\left(\omega^2-\omega_0^2\right)^2+\frac{\omega^2}{\tau^2}} \qquad (2)$$

$$P(\omega) = \frac{1}{2}E_0^2\,\frac{\varepsilon_0\omega_0^2}{\tau}\,\frac{\omega^2}{\left(\omega^2-\omega_0^2\right)^2+\frac{\omega^2}{\tau^2}} \xrightarrow[\tau\to 0]{} \frac{1}{2}E_0^2\,\varepsilon_0\omega_0^2\tau$$

Lorentzian shape

It is very useful to analyze $\frac{1}{\tau} \to 0$ limit (ballistic motion)

Then $P(\omega) = \frac{1}{2} E_0^2 \varepsilon_0 \omega_0^2 \dfrac{\frac{1}{\tau}}{\frac{1}{\tau^2} + \frac{(\omega^2-\omega_0^2)^2}{\omega^2}} \underset{\frac{1}{\tau}\to 0}{\Rightarrow}$

$$\Rightarrow \frac{1}{2} E_0^2 \varepsilon_0 \omega_0^2 \, \pi \, \delta\left(\frac{\omega^2-\omega_0^2}{\omega}\right)$$

because of identity $\frac{1}{\pi} \frac{\epsilon}{x^2+\epsilon^2} \xrightarrow[\epsilon\to 0]{} \delta(x)$

$$\delta\left(\frac{\omega^2-\omega_0^2}{\omega}\right) = \delta\left(\frac{(\omega-\omega_0)(\omega+\omega_0)}{\omega}\right) \simeq \frac{1}{2}\delta(\omega-\omega_0) + \frac{1}{2}\delta(\omega+\omega_0)$$

Absorbed power has resonance at ω_0 = plasma frequency.

<u>If we</u> neglect charge build-up, which from (1) amounts to $\vec{E}_{TOT} = \vec{E}_0$.

By eq. (2) this is the limit of $\omega_0 \to 0$ so that $\dfrac{\omega(\omega \pm \frac{i}{\tau})}{\omega(\omega \pm \frac{i}{\tau}) - \omega_0^2} \to 1$

we obtain $P(\omega) = \frac{1}{2} E_0^2 \frac{\varepsilon_0 \omega_0^2}{\tau} \dfrac{\omega^2}{\omega^4 + \frac{\omega^2}{\tau^2}} = \frac{1}{2} E_0^2 \frac{\varepsilon_0 \omega_0^2}{\tau} \dfrac{1}{\omega^2 + \frac{1}{\tau^2}}$

Observe that now $\dfrac{1/\tau}{\omega^2 + 1/\tau^2} \xrightarrow[\frac{1}{\tau}\to 0]{} \pi\,\delta(\omega)$,

i.e. peak of absorption is now at $\omega = 0$ frequency!